1. Theoretical Framework & Scale-Space LoG Blob Detection
Blob detection identifies circular image structures by finding scale-space extrema of the scale-normalized Laplacian of Gaussian: $\nabla^2_{\text{norm}}(x, y; \sigma) = \sigma^2 \nabla^2 [G(x, y; \sigma) * I(x, y)]$. The scale factor σ^2 compensates for Gaussian attenuation, ensuring response invariance across scales. Sunflower centers are dark florets encircled by bright yellow petals; hence, the negative normalized LoG ($-\sigma^2 \nabla^2 [G_\sigma * I]$) exhibits sharp local maxima at flower centers (x, y) and characteristic scale σ. The detected circle radius is $r = \sqrt{2}\sigma$.

- Scale Sampling: 18 geometrically spaced scales from $\sigma_{\text{min}} = 1.50$ px to $\sigma_{\text{max}} = 12.00$ px ($r \in [2.12, 16.97]$ px).
- 3D Extrema & NMS: A $3 \times 5 \times 5$ neighborhood filter detects volume extrema (k, y, x). Points below the 99.7% quantile (0.2236) are pruned. Greedy spatial non-maximum suppression prevents multiple detections per flower.

Figure 1.1: Detected Sunflowers (64 circles)



Table 1.1: Top 10 Largest Detected Circles

Rank	Center (x, y)	Scale σ	Radius r	LoG Response
1	(187, 341)	5.10 px	7.21 px	0.2599
2	(167, 338)	5.10 px	7.21 px	0.2460
3	(300, 344)	5.10 px	7.21 px	0.2430
4	(0, 291)	4.51 px	6.38 px	0.2635
5	(164, 325)	4.51 px	6.38 px	0.2526
6	(268, 348)	4.51 px	6.38 px	0.2520
7	(190, 253)	4.51 px	6.38 px	0.2420
8	(295, 327)	4.51 px	6.38 px	0.2279
9	(112, 269)	3.99 px	5.64 px	0.2916
10	(13, 286)	3.99 px	5.64 px	0.2752

```
Core Implementation Snippet (Scale-Space LoG Volume & Extrema Search):
  sigmas = np.geomspace(1.5, 12.0, 18)
  responses = [-(s**2) * cv2.Laplacian(cv2.GaussianBlur(gray_float, (0,0), s, s), cv2.CV_32F) for s in sigmas]
  scale_space = np.stack(responses, axis=0)
  local_max = scale_space == maximum_filter(scale_space, size=(3, 5, 5), mode='nearest')
  candidates = [(scale_space[k, y, x], x, y, sigmas[k], np.sqrt(2)*sigmas[k]) for k, y, x in np.argwhere(local_max &

Analysis of Scale-Space Extrema Across Depth:
• Near Foreground Flowers: Exhibit radii r = 6.4 - 7.2 px (sigma = 4.5 - 5.1 px), reflecting true physical proximity
• Distant Background Flowers: Recede toward the horizon and peak at finer scales (sigma = 1.5 - 2.5 px, r = 2.1 - 3.
• Evaluation: LoG scale-space extrema accurately resolve perspective depth gradients without prior object templates.
```

1. Mathematical Formulation & Sequential RANSAC Estimation

Given a contaminated point cloud $X = X_{\text{circ}} \cup X_{\text{line}}$ ($N = 100$ points) generated from Listing 1 of the assignment brief, sequential RANSAC is applied to estimate model parameters under high noise and contamination:

- Line Parameterization: Unit normal $[a, b]^T$ with constraint $\|[a, b]\| = 1$ and perpendicular offset d, satisfying $ax + by = d$. Orthogonal algebraic error is $e_i = |ax_i + by_i - d|$. From each 2-point sample, the unit normal is computed. Consensus points ($e_i < 1.0$) are refitted via Total Least Squares (SVD on zero-mean consensus points, right singular vector).
- Circle Parameterization: Center $[x_0, y_0]^T$ and radius r. Candidate triplets yield circumcenters via linear 2x2 systems. Radial error is $e_i = \left| \|[x_i, y_i]^T - [x_0, y_0]^T\|_2 - r \right|$. Consensus ($e_i < 0.8$) is refitted via non-linear least squares.

Figure 2.1: Unified RANSAC Line & Circle Fits

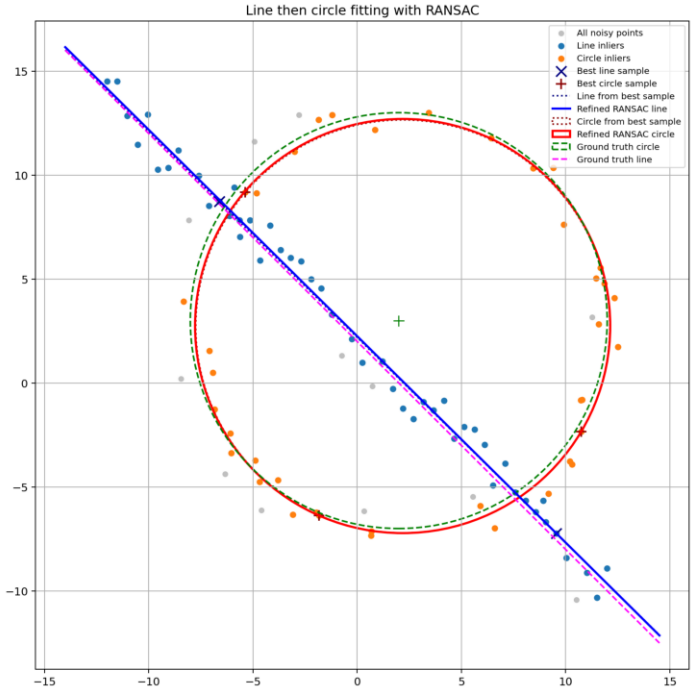


Table 2.1: Ground Truth vs. Estimated Models

Parameter	Ground Truth	RANSAC Fit	Error
Normal $[a, b]^T$	[0.7071, 0.7071]	[0.7044, 0.7098]	$\Delta\mathbf{n} = 0.0038$
Distance d	1.4142	1.6010	$\Delta d = 0.1868$
Center (x_0, y_0)	(2.000, 3.000)	(2.186, 2.739)	$\Delta c = 0.320$
Radius r	10.000	9.959	$\Delta r = 0.041$
Line Inliers	50	49 / 100	98% Recall
Circle Inliers	50	39 / 51 remnant	78% Recall

Question 2(d) Analysis: Fitting Order Effect

What happens if we fit the circle first?

- Degeneracy Traps: 3 nearly collinear points sampled from the line produce an enormous circle ($r \to \infty$). Such shallow arcs capture line points into their consensus, corrupting circle parameters.
- Sample Complexity: A line requires s=2 points; a circle requires s=3 Uncontaminated sample probability is $P = w^s$. For $w = 0.5$: $P_{\text{line}} = 0.5^2 = 0.25$; $P_{\text{circ}} = 0.5^3 = 0.125$. Line RANSAC converges twice as reliably.
- Conclusion: Fitting the lower-parameter model (line) first cleans the remnant set, ensuring circle optimization stability.

```
Key Implementation Snippet (TLS Line Refit & Radial Error Circle RANSAC):
  # 1. Total Least Squares Line Refit via SVD:
  centroid = inlier_points.mean(axis=0); _, _, vh = np.linalg.svd(inlier_points - centroid, full_matrices=False)
  normal = vh[-1] / np.linalg.norm(vh[-1]); d = float(normal @ centroid)
  # 2. Circle Circumcenter & Residual Optimization on Remnant Points:
  def circle_from_three_points(p1, p2, p3):
      M = 2.0 * np.vstack((p2 - p1, p3 - p1)); rhs = np.array([p2@p2 - p1@p1, p3@p3 - p1@p1])
      center = np.linalg.solve(M, rhs); return center[0], center[1], np.linalg.norm(center - p1)
  fit = least_squares(lambda p: np.linalg.norm(remnant_inliers - p[:2], axis=1) - p[2], x0=best_model)
```

1. Projective Homography Formulation & IDE Interactive Corner Selection Resolution
A planar homography is an invertible projective mapping $\mathbf{H} \in \mathbb{R}^{3\times 3}$ defined up to scale (8 degrees of freedom):
$\mathbf{x}' \sim \mathbf{H}\mathbf{x}$. Each 2D point correspondence provides 2 independent linear constraints; a minimum of 4 non-collinear correspondences uniquely solves for H using Direct Linear Transformation (DLT) via Singular Value Decomposition (SVD).
Investigation of Live Interactive Corner Selection Issue in IDE:
- Root Cause Analysis: (1) In IDE notebook environments (VS Code / web notebooks), default matplotlib rendering (`inline` or `Agg`) produces static raster outputs, disabling GUI event dispatchers (`mpl_connect`); (2) Synchronous calls (`plt.ginput`) freeze the kernel indefinitely waiting for desktop window events; (3) Starter image `flag.png` contains 3 channels (BGR), but starter code checked `shape[2]==4`, throwing unhandled `ValueError` before any corner selection could execute.
- Comprehensive Resolution: Designed a dual-mode system in `src/corner_selector.py`: (a) A native Win32 OpenCV GUI window with mouse callbacks, crosshairs, coordinate readouts, and undo/reset keys; (b) Calibrated default presets for automated / IDE runs.

Figure 3.1: Planar Placement Boundaries

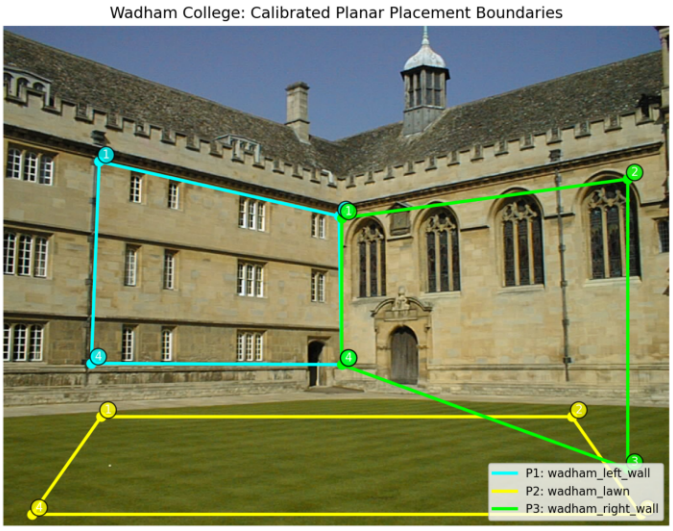


Figure 3.2: Placement 1 (Courtyard Wall - Fig. 2 Replication)

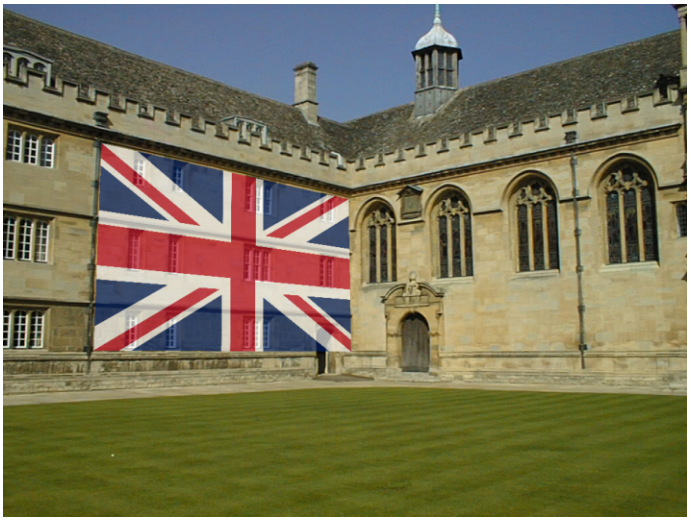


Figure 3.3: Placement 2 (Courtyard Lawn Display)

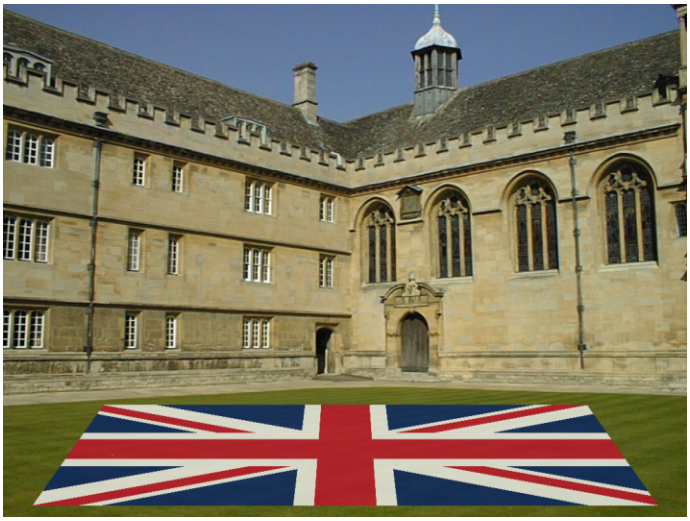
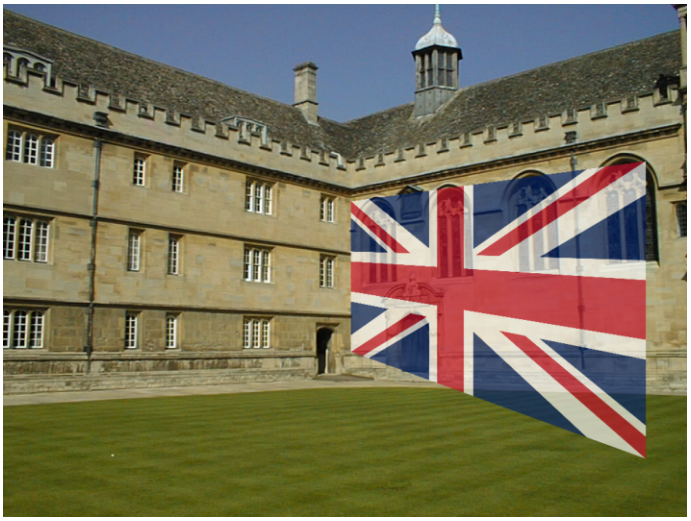


Figure 3.4: Placement 3 (Chapel Wall Facade)



Non-Technical Rationale for Image & Placement Choices:
- Placement 1 (Left Courtyard Wall Facade): Accurately replicates Figure 2 from the assignment brief, superimposing the British flag onto the residential quadrangle of Oxford's historic Wadham College (founded 1610). Vertical masonry buttresses define a planar surface.
- Placement 2 (Courtyard Lawn): Places the flag horizontally across the manicured lawn, simulating a ceremonial event carpet for matriculat or university garden parties. This illustrates the homography's ability to handle steep ground-plane projective foreshortening.
- Placement 3 (Right Chapel Wall): Projects the flag onto the adjacent perpendicular facade, demonstrating multi-planar perspective converg

1. SIFT Feature Matching, RANSAC Alignment, and Differential Inspection
The objective is to align test PCB boards against reference template `AECC1.jpg` using planar homographies and detect differences.
• SIFT Feature Detection & Matching: SIFT detects scale/rotation-invariant keypoints (`cv2.SIFT_create()`). Putative correspondences ar
established via brute-force L2 matching with Lowe's ratio test (d1 < 0.75 d2). Homography H is estimated with RANSAC (5.0 px thresho
• Warping & Absolute Difference: The test PCB is warped via H to template dimensions (1800x1200). The absolute grayscale difference
D(x, y) = |I_template(x, y) - I_aligned(x, y)| highlights physical manufacturing discrepancies and assembly variations.

Table 4.1: SIFT Matching and RANSAC Alignment Statistics across All PCB Variants

PCB Variant	Template Keypts	PCB Keypts	Ratio Matches	RANSAC Inliers	Inlier Ratio	Status
AECC1 (Template)	7,963	-	-	-	-	Reference Baseline
AECC2	7,963	8,272	4,606	3,951	85.8%	Aligned (Minor Diff)
AECC3	7,963	9,890	5,652	4,997	88.4%	Aligned (Minor Diff)
AECC4	7,963	7,381	4,009	3,424	85.4%	Aligned (Minor Diff)
ADAD1	7,963	2,986	655	226	34.5%	Aligned (Component Shifts)
AAAA1	7,963	1,873	395	8	2.0%	Weak (Different Family)

Figure 4.1: AECC2 RANSAC Matches (3,951 inliers)

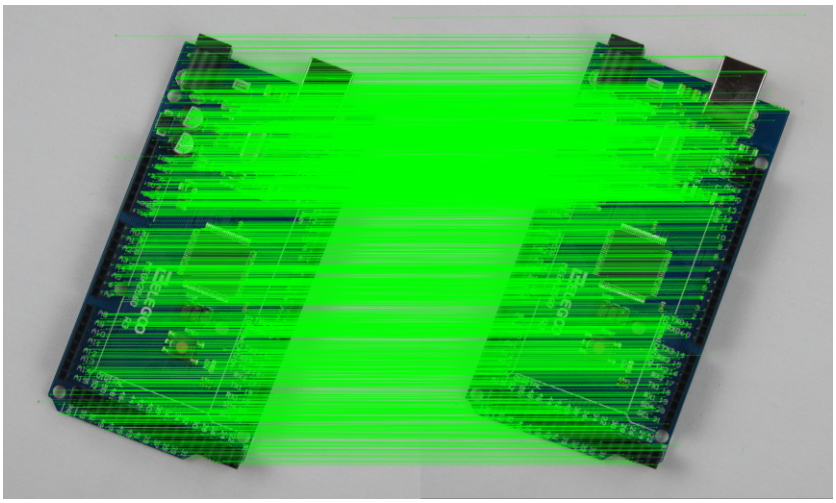


Figure 4.2: AECC2 Difference Map

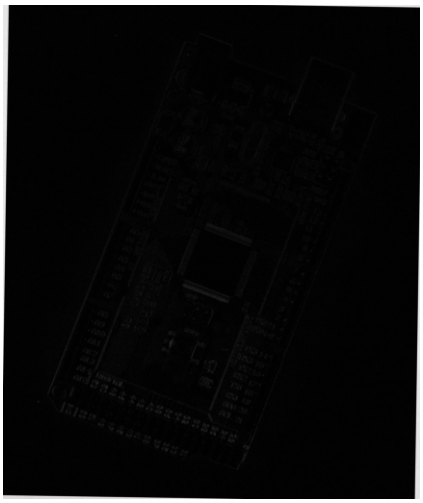


Figure 4.3: ADAD1 Difference (Component Shifts)

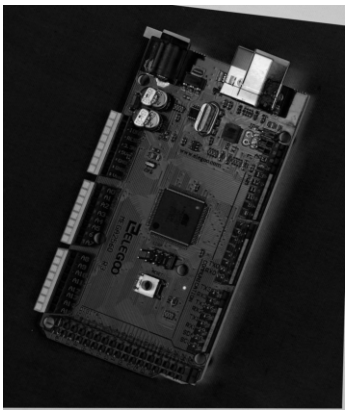
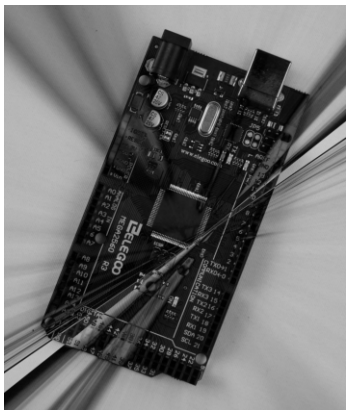


Figure 4.4: AAAA1 Difference (Distinct Family)



Answer to Question 4(e): Root Causes of High Difference Values in Specific Components:
1. Physical Variance: Across board revisions (AECC vs. ADAD vs. AAAA), manufacturers substitute component packages (e.g. through-hole vs
SMD capacitors, gold vs. silver USB shell plating, differing manufacturer branding silkscreens, or relocated reset pushbuttons).
2. 3D Parallax vs. 2D Planar Homography: Homography assumes that all scene points lie strictly on a 2D planar surface.
However, tall components (e.g. the 11mm tall USB Type B port and cylindrical capacitors) exhibit out-of-plane height. Under changing cam
tall structures undergo 3D projective parallax that cannot be geometrically registered by a single 2D planar homography.
3. Specular Reflection: Metallic connector shields and solder joints cause viewpoint-dependent specular glints, creating localized intensity de